Suppose if the work is done on the system, then $V_i > V_f$. Then, W is negative.
Note here the pressure P is inside the integral in equation (8.17). It implies that while the system is doing work, the pressure need not be constant. To evaluate the integration we need to first express the pressure as a function of volume and temperature using the equation of state.

8.6.5 PV diagram

PV diagram is a graph between pressure P and volume V of the system. The P-V diagram is used to calculate the amount of work done by the gas during expansion or on the gas during compression. In Unit 2, we have seen that the area under the curve will give integration of the function from lower limit to upper limit. The area under the PV diagram will give the work done during expansion or compression which is shown in Figure 8.22

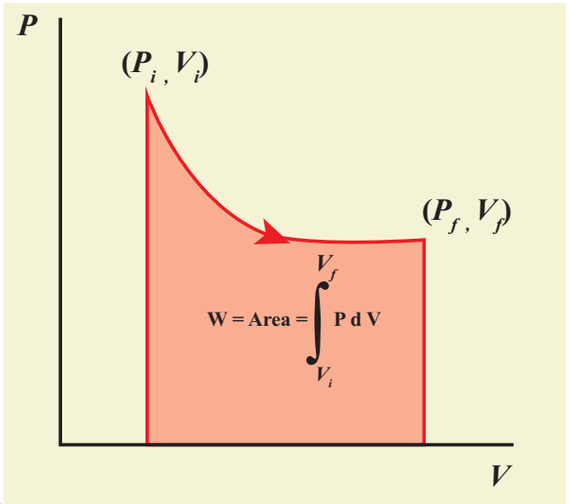


Figure 8.22 Work done by the gas during expansion

The shape of PV diagram depends on the nature of the thermodynamic process.

EXAMPLE 8.15

A gas expands from volume $1m^3$ to $2m^3$ at constant atmospheric pressure.

(a) Calculate the work done by the gas.

(b) Represent the work done in PV diagram.

Solution

(a) The pressure P = 1 atm = 101 kPa, $V_f = 2\ m^3$ and $V_i = 1m^3$

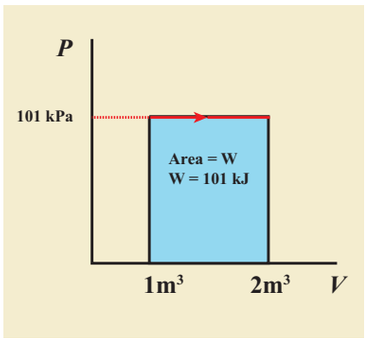
From equation (8.17)

$$W = \int_{V_i}^{V_f} PdV = P\int_{V_i}^{V_f} dV$$

Since P is constant. It is taken out of the integral.

$$W = P\,(V_f - V_i) = 101\times10^3 \times (2 - 1) = 101\ kJ$$

(b) Since the pressure is kept constant, PV diagram is straight line as shown in the figure. The area is equal to work done by the gas.



Note the arrow mark in the curve. Suppose the work is done on the system, then volume will decreases and the arrow will point in the opposite direction.

8.7 SPECIFIC HEAT CAPACITY OF A GAS

Specific heat capacity of a given system plays a very important role in determining

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